



Achieving tunable perfect microwave absorption through a reconfigurable cementitious metastructure integrating flexible carbon nanotube frequency selective surface

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ABSTRACT

Given the increasingly intricate electromagnetic (EM) environments and applications, deformable and reconfigurable absorbers have been extensively developed for tunable EM wave absorption. However, those reconfigurable absorbers are usually sensitive to external force disturbances, affecting their absorption performance and mechanical stability. To overcome this challenge, this study proposes a novel reconfigurable metastructure, simultaneously ensuring tunable absorption and performance stability. The metastructure consists of a 3D cementitious metamaterial substrate (CMS) and a reconfigurable carbon nanotube-frequency selective surface (CNT-FSS) device. The CMS is designed to modulate and aggregate EM waves of various frequencies to different targeting positions, and the CNT-FSS device is integrated at these positions to absorb the aggregated EM waves at specific frequencies. The CNT-FSS device is fabricated by coating CNT solution onto a flexible fiber fabric, forming an impedance gradient structure with enhanced loss characteristics. As a proof of concept, a reconfigurable metastructure is manufactured, enabling convenient switching between adhering or suspending the CNT-FSS device at various positions on the CMS. Experimental results prove that the performance of the metastructure switches between broadband absorption (reflectivity below -10 dB in 5.8–18 GHz), perfect X band absorption (-22.6 dB average reflectivity and -54 dB $RL_{\min}$), and perfect Ku band absorption (-24.1 dB average reflectivity and -57.9 dB $RL_{\min}$). The findings not only significantly enhance and enrich the performance of the cementitious absorbers, but also promote an impressive approach to dynamically control EM wave absorption for diverse scenarios including broadband wave absorption, dynamic EM energy harvesting, and highly sensitive sensors.

1. Introduction

The electromagnetic (EM) wave absorption characteristics are crucial for modern engineering facilities, such as reducing radar cross-section (RCS) [1], achieving EM interference (EMI) shielding [2,3] and protecting human body from health risks caused by radiation [4,5]. Cementitious composites, as the most extensively used construction material worldwide, have emerged as excellent candidate for achieving both mechanical and absorption functionalities due to their inherent EM loss characteristics and adjustable performance. In the past years, a great deal of effort has been exerted to construct cementitious material absorbers to achieve low reflectivity and broadband absorption [6–11]. Recently, the structural design of cementitious absorbers has attracted considerable attention, such as multilayer structures [12–15], metamaterials [16,17] and metastructures [18,19] to reach a balance of the

impedance matching and dissipating capacity of absorbers. For example, Yang et al. proposed a novel 3D porous periodic array structural foam cement-based absorbing metamaterials (FCAM), demonstrating an ultra-broadband ($RL < -20$ dB) of 13.85 GHz and absorbing peak of -56.59 dB in 2–18 GHz [17]. Excellent lightweight and broadband absorption performance can be achieved by foamed cement-based composites incorporating carbon materials [12,20]. However, conventional absorbers are dispersive and suffer from uncontrollable and unchangeable functions including absorption efficient and working frequencies. Given the increasingly intricate EM environments and applications, absorbers are required to dynamically control their absorption performance to broaden their working frequency, facilitate functional switching, and respond to diverse EM scenarios [21].

Over the past years, significant endeavors have been directed towards reconfigurable devices aimed at tunable EM performances, for

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instance, tuning EM properties via lumped capacitors [22], diodes [23] and coded metamaterials [24,25], controlling absorption efficiency by mechanical compression [26,27], external voltage [28] or electron irradiation [29], tuning absorption frequency utilizing shape-change materials [30–33] and reconfigurable frequency selective surfaces (FSSs) [34–36]. Notably, among all the proposed methods to dynamically control EM waves, reconfigurable devices based on shape-change manners have great potentials to achieve tunable EM properties, as the transition from 2D structures to 3D counterparts may yield dramatic changes in EM responses [37]. For example, an accordion-origami metamaterial absorber can realize the switching between ultra broadband absorption and high reflection through deformation [38]. Reconfigurable manners have also been employed to dynamically control the broadband absorption [39] and wide-angle absorption performances [40]. Absorption bandwidths can be mechanically switched to cover the S, C, X and Ku bands by rotating kirigami-pixels in predestinated orientations [39]. Wide-angle EM absorption can be achieved in a dynamically controllable manner via a reconfigurable origami-based metamaterial [40]. However, reconfigurable devices through shape-change manners in many cases leads to an unstable deployed state, i.e., small perturbations may collapse the structure along the same deployment path [41]. Therefore, achieving mechanical stability, especially under multi-directional loads, poses a significant challenge for shape-changing reconfigurable absorbers. Furthermore, when applied to large-scale infrastructure and engineering, these reconfigurable methods might face limitations considering the requirements for robustness, durability, and compatibility.

To address the challenge of mechanical stability and practical applications, this study proposes a novel reconfigurable metastructure, consisting of a cementitious metastructure substrate (CMS) with 3D gratings and a reconfigurable carbon nanotube-frequency selective surface (CNT-FSS) device. Comparative experiments and numerical simulations are conducted to thoroughly analyze the absorption mechanisms of CMS. Leveraging the mechanisms of CMS in regulating and aggregating EM energy across various frequency bands, the CNT-FSS devices are integrated to specific positions to absorb the aggregated EM waves. As a proof of concept, a reconfigurable metastructure is manufactured by a CNT-FSS device applying CNT film onto a flexible and transparent fiber fabric, enabling convenient switching between adhering or suspending of the CNT film at various positions on the CMS. The roles of the CNTs and the fiber fabric, and the mechanisms of the device for modulating EM absorption, are analyzed in detail. Finally, the absorption performance and tunability of the reconfigurable metastructures are experimentally validated.

2. Methods

2.1. Theoretical design principles

2.1.1. Theoretical analysis

According to the transmission line method, the EM wave absorption performance of an absorber is expressed in terms of reflection loss (RL) as follows [17,42]:

$$RL = 20 \lg \left| \frac{Z_{in} - Z_0}{Z_{in} + Z_0} \right| \quad (1)$$

where, Z_0 is the wave impedance of free space, which is taken as 377Ω , and Z_{in} is the input impedance of the absorber, which can be expressed as follows [17,42]:

$$Z_{in} = Z_0 \sqrt{\frac{\mu_r}{\epsilon_r} \tanh \left(j \frac{2\pi f}{c} d \sqrt{\epsilon_r \mu_r} \right)} \quad (2)$$

where d is the thickness of the absorber, f is the frequency of the incident wave, c is the speed of light, ϵ_r and μ_r represent the complex relative permittivity and permeability of the absorber material. The EM wave

attenuation capability of the absorber is influenced by the relative complex permittivity $\epsilon_r = \epsilon' - j\epsilon''$ and relative complex permeability $\mu_r = \mu' - j\mu''$. Therefore, the traditional method to enhance the absorber's performance mainly involves improving the natural ϵ_r and μ_r of the material to enhance its absorptive properties. According to Eq. (1), when RL is close to zero, Z_{in} should approach Z_0 :

$$Z_{in} = Z_0 \sqrt{\frac{\mu_r}{\epsilon_r}} Z_0 \quad (3)$$

Indicates that $\epsilon_r \rightarrow \mu_r$. Therefore, to achieve the maximum absorption, the EM parameters of the material should be matched with natural space as much as possible, and relative permittivity and permeability of the absorber should be matched with each other.

On the other hand, enhancing the loss characteristics of the absorber is also crucial for improving absorption performance. The loss characteristics of absorbers are typically evaluated using the loss coefficient (α), expressed as follows [20]:

$$\alpha = \frac{\sqrt{2\pi}f}{c} \sqrt{(\mu''\epsilon'' - \mu'\epsilon') + \sqrt{(\mu''\epsilon'' - \mu'\epsilon')^2 + (\mu'\epsilon'' + \mu''\epsilon')^2}} \quad (4)$$

Moreover, introducing lossy materials, increasing the effective path length and utilizing the resonant absorption design are effective methods for enhancing loss characteristics.

2.1.2. Design of the CMS

In order to optimize the input impedance and enhance the EM wave absorption efficiency of cementitious materials, this study proposes a cementitious metastructure substrate (CMS) by constructing millimeter-scale gratings on the surface of the cementitious slab. This structure can exhibit broadband absorption performance by optimizing the equivalent Z_{in} , increasing the interaction between EM waves and the structure, and enlarging the effective path length. Cementitious materials generally have a larger ϵ_r than μ_r , which is not conducive to impedance matching condition in Eq. (3). The design of metastructures can effectively reduce the equivalent relative permittivity (ϵ_{eff}) to improve impedance matching characteristics. For an infinitely large grating structure, it can be assumed that the material is uniformly distributed in the air. The ϵ_{eff} can be calculated using the Maxwell Garnett model:

$$\frac{\epsilon_{eff} - \epsilon_h}{\epsilon_{eff} + 2\epsilon_h} = f \frac{\epsilon_n - \epsilon_h}{\epsilon_n + 2\epsilon_h} \quad (5)$$

In which ϵ_h and ϵ_n are the relative dielectric constants of air and cementitious material, respectively, and f is the proportion of cementitious material. Therefore, impedance matching performance of the CMS can be improved by optimizing the proportion of the grid. To further enhance the absorption characteristics and achieve dynamic control of EM absorption, based on the absorption mechanisms analyses of the CMS and the distribution of EM energy in the structure, this work systematically designs a reconfigurable CNT-FSS and proposes a simple preparation method to integrate the FSS with CMS, which will be introduced and investigated in detail in subsequent sections.

2.2. Materials

The following materials are used to prepare the CMS: P.O. 52.5 cement, ultra-fine sand, fly ash (FA) and polyvinyl alcohol (PVA) fiber with a diameter of $40 \mu\text{m}$ and a length of 9 mm. For the CNT-FSS, the

Table 1
Basic properties of the carbon nanotubes.

CNTs	Diameter (nm)	Length (μm)	Purity (wt %)	Specific surface area (m^2g^{-1})
Multiwall CNT	20–25	3–10	$\geq 97.5 \%$	170–250

basic properties of CNTs are listed in Table 1. Flexible and transparent silicon-aluminum ceramic fiber fabric is utilized as the substrate of CNTs.

2.3. Fabrication

The mixing proportions of the CMS are listed in Table 2. To fabricate the CMS, the ultra-fine sand, cement, and fly ash are stirred in the mixer at 107 rpm for 2 min. The mixing water is poured into the mixer at 107 rpm for 4 min. Then the mixing speed is increased to 365 rpm for another 2 min. Finally, PVA fibers are added and stirred for another 2 min at 365 rpm. The fresh mixture is poured into molds measuring $180 \times 180 \times 25$ mm.

As shown in Fig. 1 (a), the preparation method of the CMS is as follows: a striped top mold with dimensions of $180 \times 180 \times 6$ mm is used, with baffles at the edges for fixing the mold into the paste. The stripe width and spacing of the top mold are both 12 mm [18]. Before the initial setting of the cement slurry, fully sink the striped mold with lubricant into the slurry and press out a groove with a depth of 6 mm. Clean up the extruded cement slurry and smooth the surface. After curing for 24 h, the specimens are demolded and grating surface structures are obtained. The width and spacing of the grating are 12 mm, and the thickness of the grating is 6 mm. Then the specimens are cured in a water bath at room temperature for 28 days. The specimens are dried in an oven at 60°C for 24 h before the testing is conducted.

For the CNT solution, polyvinyl pyrrolidone (PVP) with 1/4 CNT weight is used as a dispersing agent. First, 93.75 % water: 5 % CNT: 1.25 % PVP are mixed using a high-speed mixer at 3000 rpm for 30 min. Then the CNT suspension is ground at 2000 rpm for 120 min by a grinder to obtain uniformly dispersed CNT conductive paste. As shown in Fig. 1 (b), the resulting CNT solution is used as the ink to be coated on the fiber fabric. The width of the periodic CNT film is 12 mm, with a CNT area density of 0.2 mg/cm^2 . After preparing the CMS and CNT-FSS device, the CNT-FSS device can be integrated to the CMS as shown in Fig. 1 (c), and the CNT-FSS can be flipped in the 3D structure to achieve dynamic control of EM absorption performance.

2.4. Characterization

2.4.1. Dielectric properties

The dielectric properties of CNTs are measured through the transmission line method in 2–18 GHz, employing a Rohde & Schwarz ZNB40 vector network analyzer. CNTs are mixed with melted paraffin wax at a volume ratio of 1:1 and prepared into a toroid-shaped sample with 7 mm in outer diameter, 3.04 mm in inner diameter and 2 mm in height. These samples are placed in a calibrated coaxial system, and the EM parameters are collected using the vector network analyzer.

2.4.2. S parameters

The transmission and reflection of the cementitious dielectric and fiber fabric in X band are measured by waveguide method employing a Rohde & Schwarz ZNB40 vector network analyzer. The size of the specimen is $10.2 \text{ mm} \times 22.8 \text{ mm} \times 3.0 \text{ mm}$. The absorption can be

Table 2
Mixing proportion of the cementitious dielectric.

Series	Cementitious binder	Ultra-fine sand	Water	CNT (wt.%)	PVA fiber (vol%)
Cement (0% CNT)	1	0.25	0.3	0	2
Cement (0.4% CNT)	1	0.25	0.3	0.4	2
Cement (1.2% CNT)	1	0.25	0.3	1.2	2

obtained from the absorption formula $A(\omega) = 1 - T(\omega) - R(\omega)$, in which the parameters $A(\omega)$, $T(\omega)$ and $R(\omega)$ denote the absorption, the transmission and the transmission, respectively.

2.4.3. Microstructures

Scanning electron microscope (SEM) tests are performed to observe the microscopic characteristics of the flexible CNT-FSS, using an FEI Quanta 650 FEG scanning electron microscope. Samples measuring $4 \times 5 \times 5$ mm were taken out from the center of specimens and were dried at 60°C for 24 h. The energy dispersion X-ray (EDX) spectrometer attached to SEM is used to determine the type and content of elements in the detected area. EDX analyses were performed using a working distance of 15.0 mm, an aperture of $60 \mu\text{m}$ and a beam voltage of 15.0 kV. The acquisition time and dwell time were set at 5 s/cycle and $100 \mu\text{s/px}$, respectively.

2.4.4. Absorbing performance

The ability to absorb EM waves is evaluated by reflectivity (unit: dB) and bandwidth (unit: GHz). The lower the reflectivity, the higher the absorption. The reflectivity of the specimens in 2–18 GHz is measured using an arch reflecting method [43], according to the Chinese national standard GJB 2038-94. Compared with the calculated reflectivity results according to measured EM parameters, the arch method, which considers the matrix and dispersion state of the absorber, reflects more reliably the actual absorbing properties of the samples [44]. The reflectivity (Γ) of the sample can be expressed as $\Gamma = 10 \lg(P_a/P_m)$, where P_a and P_m are the measured reflecting power of the sample and the reflective metal plate, respectively. Then the absorption (A) can be calculated by $A = 1 - 10^{\Gamma/10}$. For example, when $\Gamma = -10 \text{ dB}$, A can be calculated to be 90 %. The absorption bandwidth with reflectivity lower than -10 dB is widely recognized as the effective bandwidth [45–47].

The experiments are carried out in an anechoic chamber with a setup of two wideband double-ridged horn antennas connected with a vector network analyzer. The two horn antennas are mounted on the arc track. The specimens of $180 \times 180 \text{ mm}$ are put at the center of the arc track. One of the horn antennas served as the source to illuminate oblique incidence plane wave to the specimen. Another horn antenna worked as the receiver. The measurements are conducted under TE and TM polarization modes.

3. Results

3.1. Absorbing performance of the CMS

3.1.1. Experimental reflectivity results of the CMS

The absorption performance of the CMS is investigated. In our previous studies, it was found that 1.2 wt% CNTs can significantly enhance the absorption performance of cementitious composites [48]. Therefore, this study selects three proportions, 0 wt%, 0.4 wt% 1.2 wt%, for a bivariate controlled study: the two variables are whether CNT is added and whether the CMS structure is adopted. The reflectivity values of four groups of samples under TE and TM polarizations are tested using the arch method, as shown in Fig. 2 (a and b). Fig. 2 (c) exhibits the schematic of TE and TM polarized EM waves incidence and the digital photograph of the CMS.

As shown in Fig. 2 (a and b), in the low-frequency range (1–4 GHz), the absorption performance of the CMS exhibits characteristics similar to the cementitious slab, showing distinct oscillations and sharp absorption peaks due to the $\lambda/4$ interference cancellation of the structure. In the mid-to-high frequency range, the CMS demonstrates significantly enhanced absorption performance compared to the cementitious slab. In the case of TE polarization where the grating is perpendicular to the electric field, as shown in Fig. 2 (a), as the frequency increases, the reflectivity values of the CMS rapidly decrease and then increase, exhibiting absorption peaks in the range of 10–15 GHz. The CMS

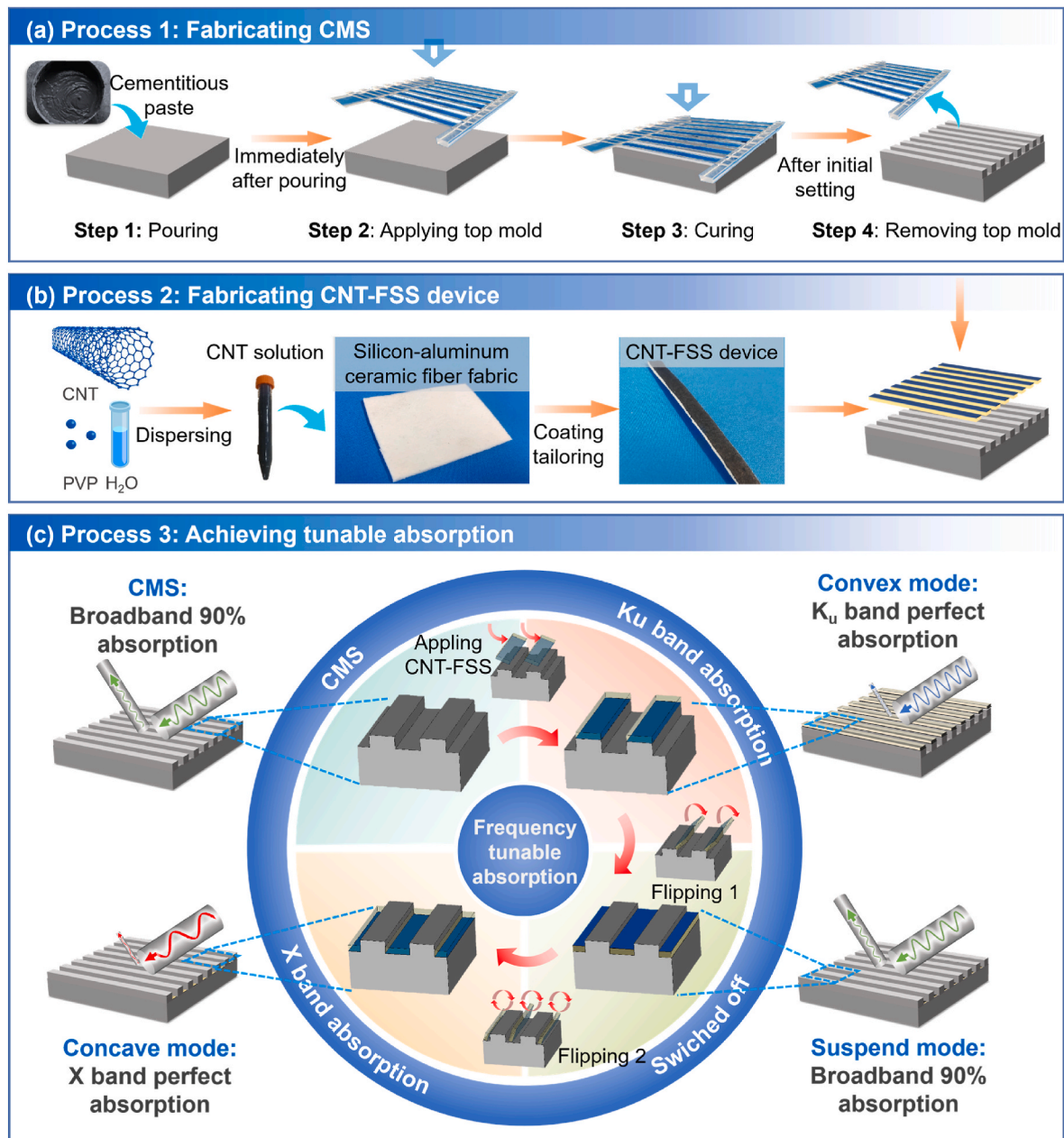


Fig. 1. Fabrication process of the CMS.

demonstrates superior broadband absorption performance: the -10 dB bandwidth reaches 11.5 GHz, and the -20 dB bandwidth reaches 5.6 GHz. Moreover, in 8.8–18.0 GHz, the reflectivity of the CMS remains below -15 dB, showing excellent absorption performance in X and K_u bands. Under TM polarization that the grating is parallel to the electric field direction, as the frequency increases, the reflectivity of the CMS gradually decreases. All of the CMSs with 0 wt%, 0.4 wt%, and 1.2 wt% CNTs contents display reflectivity peaks in the high-frequency range (16–18 GHz), measuring -20.7 dB, -21.0 dB and -22.9 dB, respectively, as shown in Fig. 2 (b). In addition, with the doping of CNT, the absorption performance in the mid-frequency range decreases.

It can be found that unlike traditional cementitious slab [48], the addition of CNTs in the CMS does not directly enhance its absorption performance. The reason is that the structure of the CMS significantly enhances its impedance matching performance from the perspective of structural design modification. However, the introduction of CNTs establishes conductive networks within the substrate (as shown in Fig. 2 (d

and e)), thereby enhancing the electrical conductivity of the material. Consequently, the impedance matching performance of the material is reduced, thus slightly reducing the overall absorption capabilities of the structure. Therefore, in the design of structured absorbers, simply incorporating absorbers into the material may not necessarily be advantageous for enhancing the overall performance of the structure. In contrast, it is essential to thoroughly investigate and optimize the impedance matching performance, loss characteristics, and absorption mechanisms of the structure. Moreover, the targeted selection of materials and the design of key positions for material functionality are crucial, contributing to the attainment of the optimal integrated material-structure design. Based on the above analyses, this study will systematically investigate the absorption mechanism of the CMS. Subsequently, further integrated FSS-structure designs and experiments will be specifically discussed in the following sections.

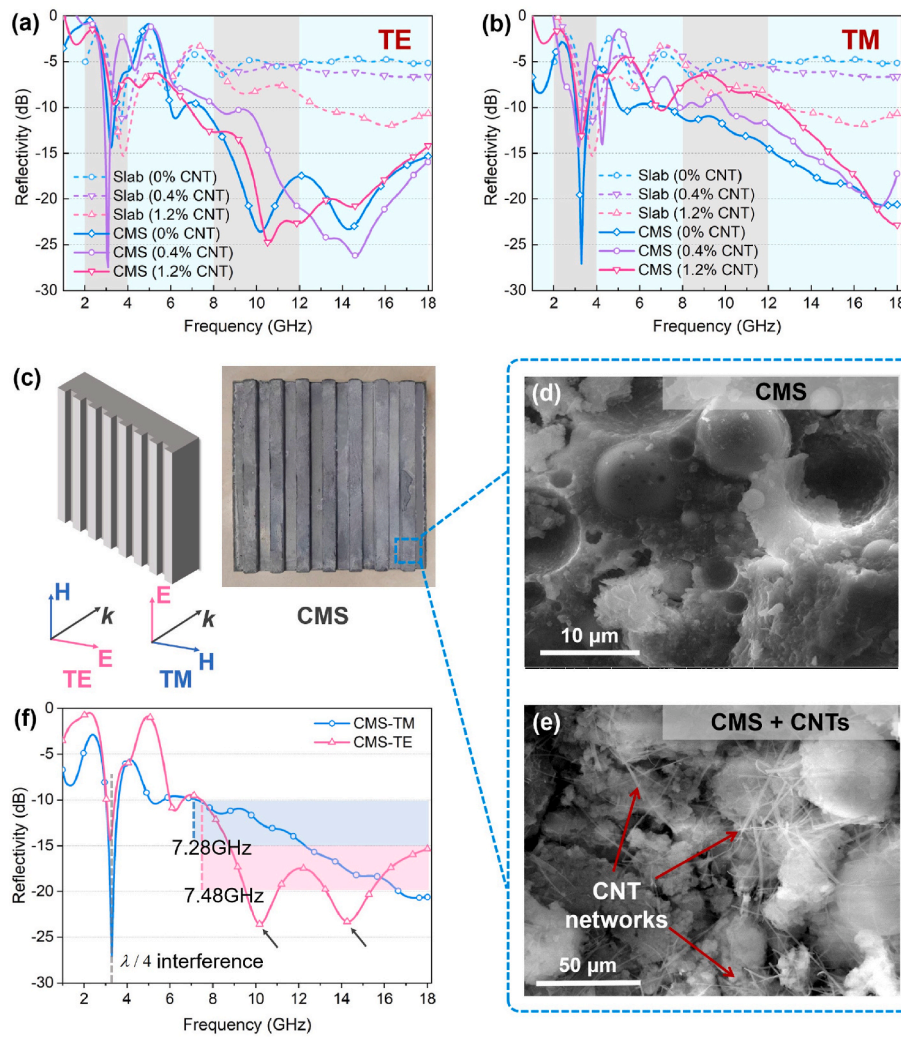


Fig. 2. (a,b) Experimental reflectivity results of the cementitious slab and CMSs with 0 wt%, 0.4 wt% and 1.2 wt% CNTs under TE and TM polarizations. (c) Schematic of TE and TM polarized EM waves incidence and the digital photograph of the CMS. (d,e) SEM images of the CMS and the CMS doping 1.2 wt% CNTs. (f) A comparison of the absorption performance of the CMS under TE and TM polarization waves.

3.1.2. Polarization sensitivity of the CMS

Due to the grating utilized in the CMS, it theoretically exhibits polarization sensitivity. Fig. 2 (f) compares the reflectivity curves of the CMS under TE and TM incidences. In the low-frequency range, reflectivity peaks appear around 3.25 GHz for both TE and TM incidences due to the $\lambda/4$ interference. As the frequency increases, the reflectivity curves under both TE and TM polarization modes show a decreasing trend. The reflectivity of the TE group steadily decreases, maintaining reflectivity below -10 dB in the 7.28–18 GHz range. It is worth noting that the reflectivity under TM polarization maintains less than -10 dB within the 7.48–18 GHz range, showing a similar effective operating frequency band to that of TE polarization. In summary, the CMS demonstrates polarization-sensitive characteristics. However, it exhibits similar operating frequency bands and excellent broadband absorption performance under both TE and TM wave incidences, with effective absorption bandwidths (EAB) reaching 11.9 GHz and 11.5 GHz, respectively, indicating its capability to achieve excellent performance simultaneously under TE and TM polarizations.

3.2. Mechanism to control EM waves of the CMS

3.2.1. Impedance matching analysis

The voltage standing wave ratio (VSWR) is a measure of impedance matching for EM wave absorbers, which can be expressed as follows

[49]:

$$VSWR = \frac{1 + \Gamma}{1 - \Gamma} \quad (6)$$

Where Γ represents the reflection coefficient. A value of VSWR close to 1 indicates a good impedance match. Fig. 3 (a) shows the VSWR curves for both the cementitious slab (with the same thickness as the CMS) and the CMS. It can be found that the value of VSMR of the CMS is significantly lower than that of the cementitious slab, especially in 4–18 GHz. As the frequency increase, the VSMR value of the CMS gradually approaches to 1, exhibiting improved impedance matching characteristics with frequency.

The impedance matching mechanism can be also discussed by Smith chart of the absorber. When the impedance curve in the Smith chart is within the -10 dB circle, it indicates that the absorber has good impedance matching performance. To further elucidate the microwave absorption mechanism of the CMS in 4–18 GHz, Fig. 3 (b) illustrates the Smith charts of CMS and cementitious slab obtained through CST Studio simulation. It can be observed that within 8–18 GHz range, the impedance matching performance of cementitious slab and CMS has significantly improved compared to that in 4–8 GHz range. The impedance curve of the cementitious slab is located outside the -10 dB circle, indicating an impedance mismatch. By modifying the surface structure

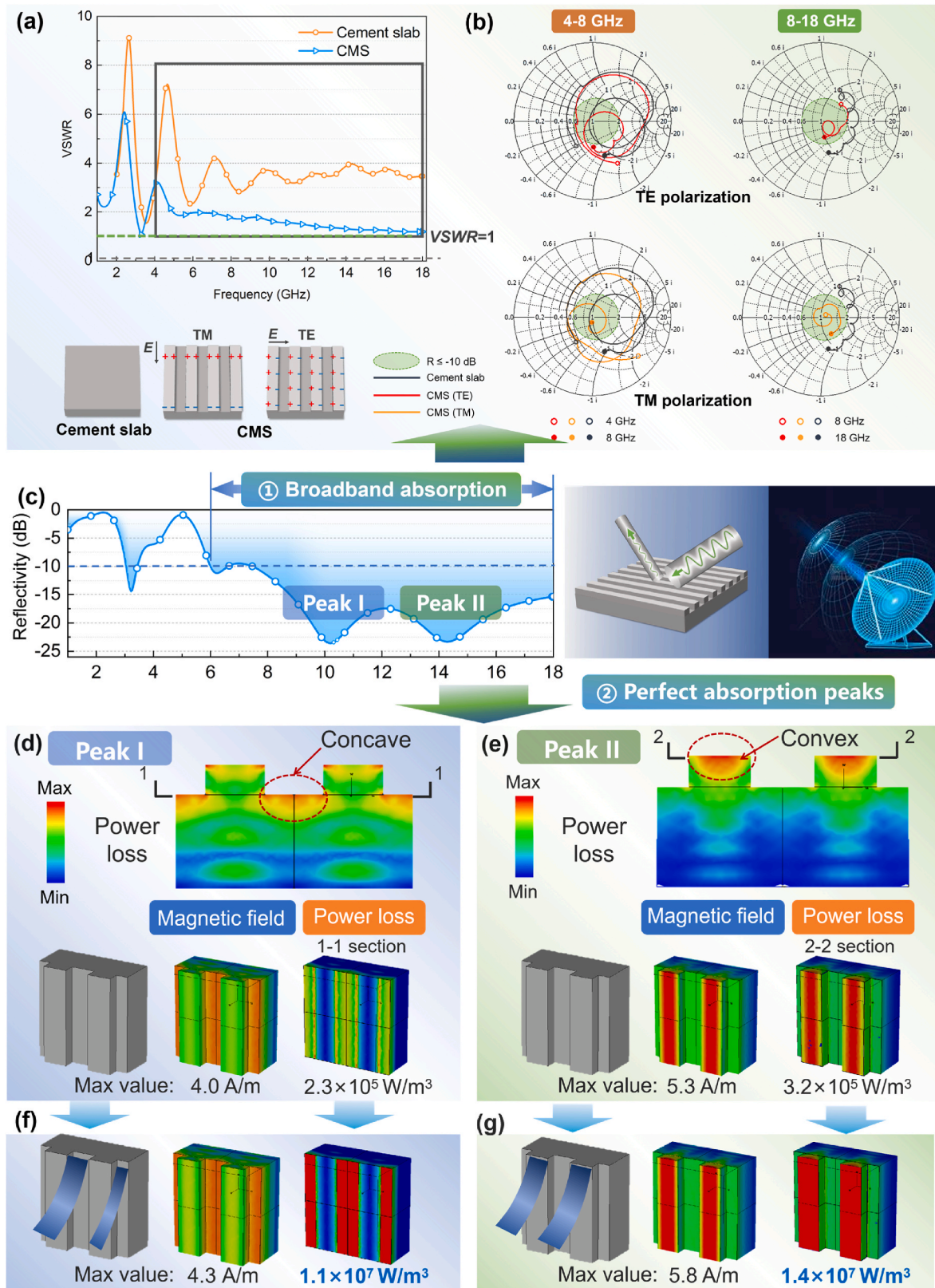


Fig. 3. (a) VSWR curves for cementitious slab and CMS. (b) The comparison of Smith charts between cementitious slab and CMS absorbers in 4–8 GHz and 8–18 GHz range. (c) Absorption performance of the CMS under TE polarization. Power loss density and magnetic field distribution of the CMS and the CMS with CNT-FSS at (d) 10 GHz and (e) 14 GHz. Magnetic field and power loss distributions of the CMS after incorporating CNT-FSS at (f) 10 GHz and (g) 14 GHz.

of the cementitious slab while using the same cementitious material, the impedance matching performance of the CMS can be significantly enhanced. Under TE polarization, the impedance of CMS in 8–18 GHz is inside of the -10 dB circle except for a few frequency bands near 8 GHz. Under TM polarization, the impedance of CMS is entirely within the -10 dB circle, indicating that CMS can significantly improve impedance matching characteristics in the X and K_u bands, achieving ideal reflectivity below -10 dB.

3.2.2. Regulating and aggregating EM energy at the peak frequencies

It can be found in Fig. 3 (c) that the reflectivity curves of the CMS exhibit several absorption peaks in various frequency ranges. In this section, absorption mechanisms of the CMS at these peak frequencies will be analyzed according to the structural characteristics of the CMS. In the low-frequency range, absorption peaks appear around 3.25 GHz for both TE and TM incidences due to the $\lambda/4$ interference. By substituting the sample thickness ($d = 25$ mm) and the corresponding EM parameters ($\epsilon' = 4.1$, $\epsilon'' = 1.0$, $\mu' = 1.65$, $\mu'' = 0.2$) measured by the transmission line method into the following interference equation [20], with $n = 3$, the calculated interference frequency is $f = 3.40$ GHz, with a 4.4 % error compared to the measured value of 3.25 GHz.

$$d = \frac{n\lambda}{4\sqrt{\epsilon_r\mu_r}} \quad (n = 1, 3, 5, \dots) \quad (7)$$

where d is thickness of the specimen, $\lambda = c/f$ is the matching wavelength at the peak.

In the mid-to-high-frequency range, under TE polarization, the CMS exhibits two absorption peaks below -20 dB around 10 GHz and 14 GHz, respectively, labeled as peaks I and II. To investigate the absorption mechanism at the peaks, this study simulates the distribution of electric and magnetic fields as well as power loss at these two peak frequencies. Fig. 3 (d) and (e) show the power loss and magnetic field distributions of the CMS at 10 GHz and 14 GHz, respectively. At 10 GHz, the power loss concentrates at the top edges and concave regions of the CMS, corresponding to the concentrated areas of the magnetic field. At 14 GHz, the power loss concentrates at the convex top of the CMS, corresponding to the concentrated area of its magnetic field. In addition, there exists power loss inside the cementitious dielectric below the 3D gratings. Due to the larger wavelength, the depth of power loss inside the dielectric is deeper at 10 GHz than that at 14 GHz. The detailed electrical and magnetic fields distributions and the absorption mechanisms can be found in Supporting Text 1 in Supporting Information.

Based on the above analysis, the absorption mechanisms of the two absorption peaks exhibited by CMS primarily stem from the intense aggregation and dissipation of EM fields generated at the concave and convex sections of the structure. This understanding advocates for the deliberate use of CNT-FSS at the targeting concave and convex locations to dissipate EM energy, making the absorber achieves perfect absorption within the targeted frequency bands. Fig. 3 (f) and (g) demonstrate the magnetic field and power loss distributions of the CMS after incorporating CNT-FSS at 10 GHz and 14 GHz, respectively. It can be found that after applying the CNT-FSS at targeting positions, the power loss value generated by the CNT film significantly increases, reaching dozens of times than that of the power loss of the metastructure without CNT-FSS. Such a substantial enhancement underscores the effectiveness of CNT-FSS in improving the performance of the metastructure absorber at specific frequencies.

3.3. Absorption mechanisms of the CNT-FSS device

3.3.1. Characterization of the CNT-FSS device

Building upon the insights gained in the previous sections, here we introduce an innovative design by integrating a reconfigurable CNT-FSS device into the CMS. As a proof of concept, the device, involving coating CNT solution onto a ceramic fiber fabric, has been designed and

fabricated, as shown in Fig. 4 (a). This device consists of 3 mm thick fiber fabric and a ~ 10 μm thick CNT film. The fiber fabric serves as a carrier for CNT film, which is flexible and wave transparent, allowing the CNT film to be conveniently adhered or equivalently suspended at various position on the CMS. In Fig. 4 (b), a cross-sectional view of the CNT-FSS device reveals that the CNT film adheres closely to the surface of the fiber fabric, with a total thickness of about 10 μm . Fig. 4 (c, d) show the elemental compositions of the fiber fabric and CNT film analyzed by EDX, with the mass and atomic percentage of the elements illustrated in the inset. It is evident that the fiber fabric primarily comprises O, Al, and Si elements. Following the application of the CNT film, the proportion of C element notably increases, accounting for 62.5 % of the total element mass. Moreover, the fiber fabric used in this study features a micro-wrinkled surface structure after applying the CNT film, as shown in Fig. 4 (e) and (f). The application of CNTs on the micro-wrinkled surface aids in enhancing multiple reflections, scattering, and interference mechanisms of the EM wave [53], further improving the absorption efficiency, as shown in Fig. 4 (j) [54,55].

3.3.2. Roles of CNTs and fiber fabric in the CNT-FSS device

The absorption mechanisms of the manufactured CNT-FSS device, including the roles of the CNTs and the fiber fabric, are analyzed in detail. Fig. 5(a–c) show the permittivity of a CNT/paraffin composite sample and the equivalent permittivity of the CNT-FSS device with the CNT film coated on the fiber fabric. CNT FSS 1 and CNT FSS 2 represent the configurations where the CNT film is located at the top and bottom of the device, respectively, as illustrated in Fig. 5 (d). As shown in Fig. 5 (a), the real part of the permittivity (ϵ') of the CNT/paraffin composite sample ranges between 5 and 7. The calculated dielectric loss tangent reaches a maximum value of 0.33 at 10.8 GHz. By adopting the CNT FSS device, ϵ' significantly decreases, and the imaginary part (ϵ'') slightly increases. This characteristic not only helps to enhance the impedance matching performance but also increases the loss tangent (ϵ''/ϵ'), thereby improving the energy dissipation capability. As shown in Fig. 5 (b), ϵ' of CNT FSS 1 decreases to a range of 2–4. When the CNT film is located at the bottom of the fiber fabric, as shown in Fig. 5 (c), ϵ' of CNT FSS 2 further decreases, mostly falling within the range of 1–1.5, making ϵ_r closer to μ_r and increasing loss tangent. Therefore, the impedance matching characteristics and the EM loss capabilities are enhanced at the CNT FSS 2 case.

Fig. 5 (e) presents the dielectric and wave-absorbing properties of the fiber fabric in device. It can be observed that ϵ' is around 1.2, and ϵ'' is close to 0, demonstrating its excellent impedance matching performance. The inset in Fig. 5 (e) shows that the reflectivity of the fiber fabric is 0 dB across the entire frequency range, proving that the fiber fabric is a wave-transparent material. Given the wave-transparent nature of the fiber fabric, why does the CNT FSS exhibit different equivalent dielectric properties depending on whether the CNT film is coated on the top or bottom surface of the fiber fabric? The following section will analyze how flipping the CNT-FSS device modulates its wave-absorbing performance.

3.3.3. Mechanisms of the CNT-FSS device for modulating EM wave absorption

As shown in Fig. 6 (a), during the CNT solution coating process, on one hand, a portion of the CNTs deposit on the surface of the fiber fabric, forming a dense CNT film. On the other hand, owing to the porous nature of the fiber fabric and the fluidity of the CNT slurry, another portion of the CNTs penetrate into the interior of the fiber fabric and coat on the fibers, forming a relatively loose conductive network with the fiber fabric. As the penetration depth of the CNTs increases, the conductive network becomes increasingly sparse until the CNT can no longer penetrate further. Through this CNT solution coating process, the CNTs and fiber fabric are fabricated into a structure with gradient impedance and progressively enhanced EM loss capability, as shown in Fig. 6 (b).

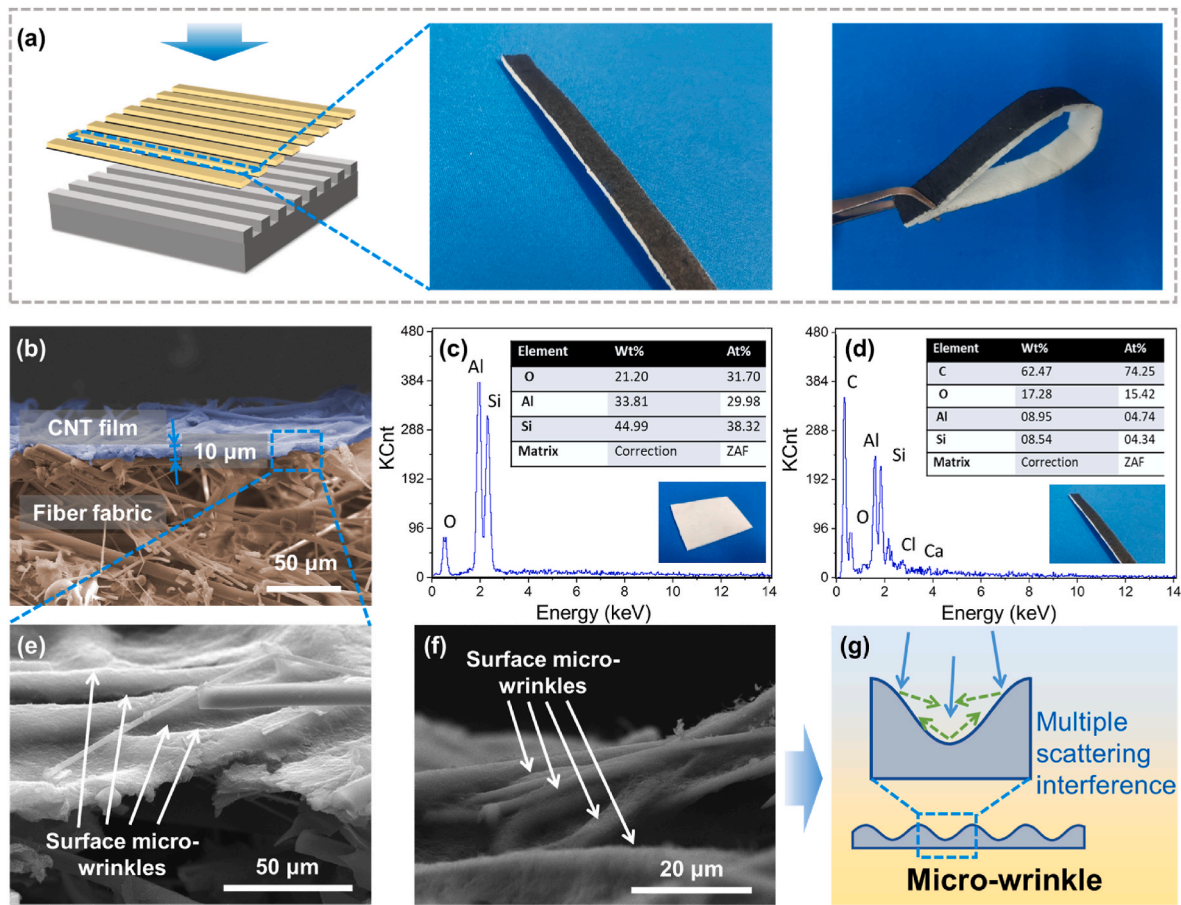


Fig. 4. (a) Schematic diagram and digital photographs of the flexible CNT-FSS device. (b) SEM image of the surface of the CNT-FSS device. (c,d) EDX results of the fiber fabric and CNT film of the CNT-FSS device. (e,f) SEM image of the surface micro-wrinkles formed by CNT coated fiber fabric. (g) Schematic diagram of the micro-wrinkle [54,55].

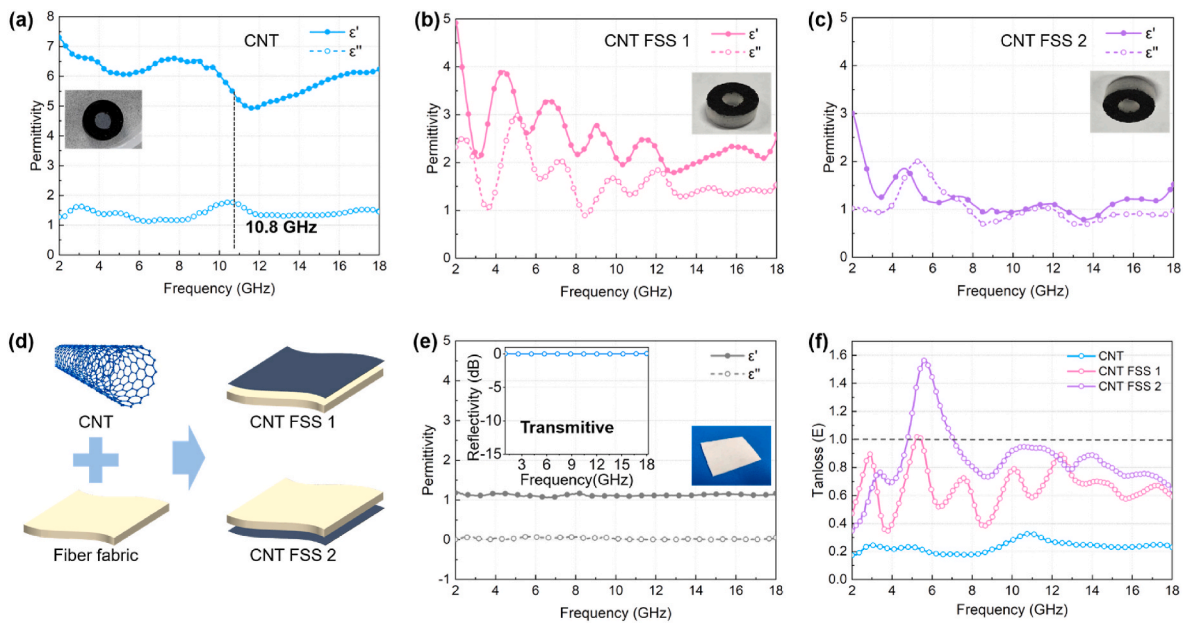


Fig. 5. (a) Permittivity of the CNT/wax composite sample. (b) Effective permittivity of the CNT FSS with the CNT film on the top. (c) Effective permittivity of the CNT FSS with the CNT film at the bottom. (d) Schematic diagram of the two kinds of CNT FSSs. (e) Permittivity of the fiber fabric. (f) Loss tangent of the different CNT samples.

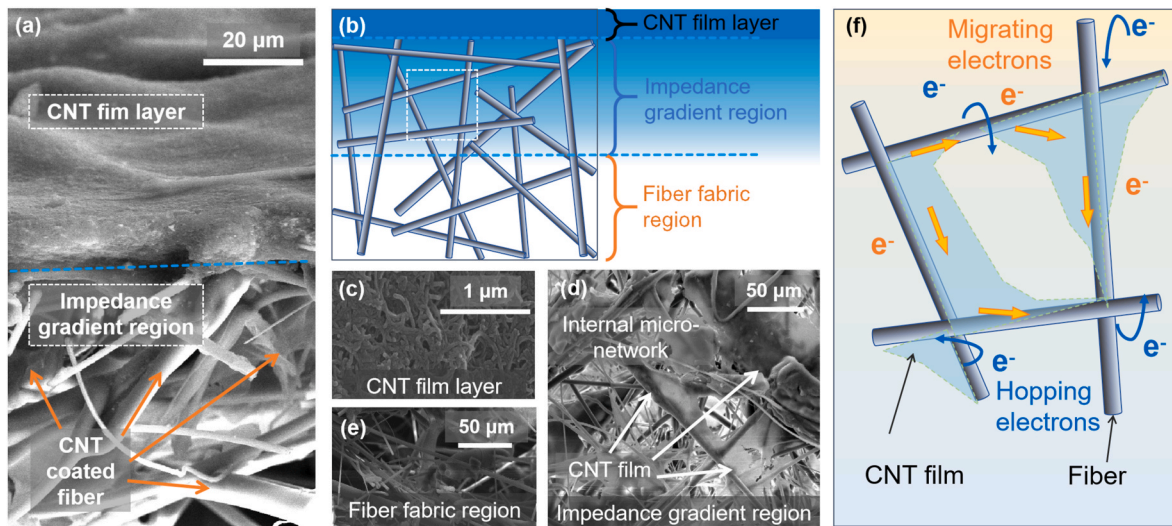


Fig. 6. (a) SEM image of the CNT film on the fiber fabric. (b) Schematic diagram of the impedance gradient structure. SEM images of the (c) CNT film layer, (d) impedance gradient region and (e) fiber fabric region. (f) Schematic diagram of the micro network in the impedance gradient region.

This impedance gradient structure includes the dense CNT film layer on the surface, the middle impedance gradient region, and the transparent fiber fabric region. Fig. 6 (c–e) exhibit the detailed SEM images of these three parts of the impedance gradient structure. As shown in Fig. 6 (d), the SEM image of the middle impedance gradient region reveals that the microstructure of the fiber fabric consists of micro networks of interlaced ceramic fibers, with CNT films coating the fiber, forming a 3D conductive network. Under the action of EM field, electrons migrate in the network to form an induced current, which is eventually converted from electrical energy to thermal energy and dissipated [56], as illustrated in Fig. 6 (f).

For CNT FSS 1, when the CNT film is located at the top of the fiber fabric, EM waves first contact with the surface CNT film. Although the CNT film possesses wave loss characteristics, its impedance mismatch with air causes the EM waves to be easily reflected. For CNT FSS 2, when the CNT film is coated on the bottom of the fiber fabric, EM waves initially pass through the fiber fabric region, which exhibits excellent impedance matching. Subsequently, the waves dissipate energy as they propagate through the intermediate impedance gradient region, and the remaining EM energy is ultimately absorbed efficiently by the CNT film at the surface. Therefore, the wave-absorbing performance is the optimum when the CNT is located at the bottom of the fiber fabric. As shown in Fig. 5 (f), the loss tangent of CNT, CNT FSS 1, and CNT FSS 2 increases sequentially. Therefore, the wave-transparent fiber fabric not only provides a flexible supporting substrate for the CNT film, allowing the CNT film to be positioned at any location in the metastructure, but also utilizes its microstructure to create an impedance gradient structure with enhanced loss characteristics, playing a significant role in improving wave-absorbing performance. As a result, the wave-absorbing capability of the CNT FSS device can be modulated by flipping its orientation.

3.4. Tunable absorption achieved by the reconfigurable metastructure

A class of reconfigurable metastructures to dynamically control EM wave absorption are proposed, as depicted in Fig. 7 (a), including adhering CNT film to the concave and convex areas of the CMS (namely concave and convex modes), and suspending CNT film within the central part of the concave area of the CMS (namely suspend mode). In both concave and convex modes, the CNT FSS adopts the configuration with the CNT film at the bottom. Reflectivity results of the proposed metastructures are shown in Fig. 7(b–f). Fig. 7 (b) demonstrates that the peaks I and II of the CMS are located at 10.23 GHz and 14.33 GHz, with

reflectivity values of -23.8 dB and -23.4 dB, respectively. Integrating CNT-FSS device can further enhance the absorption performance of the CMS. Fig. 7 (c and d) respectively present the average reflectivity, $RL_{\min}$, and -10 dB and -20 dB bandwidths of the three modes of the metastructure. The average reflectivity of the CMS, concave mode, and convex mode decreases from -13.5 dB to -14.9 dB and -16.1 dB, indicating that the average absorption of the convex mode exceeds 97 %. The -10 dB EAB of the three modes sequentially increases to 11.5 GHz, 12.4 GHz, and 12.8 GHz. The -20 dB bandwidth expands from 3.5 GHz in the CMS to 5.3 GHz in the concave mode, and further widens to 6.4 GHz in the convex mode.

Since the two peaks displayed by CMS are due to the aggregation of waves into the concave and convex parts, integrating CNT-FSS separately in these parts significantly enhances the absorption near one of the peaks. As depicted in Fig. 7 (e), the concave mode notably reduces the reflectivity of peak I, achieving $RL_{\min}$ of -54.0 dB at 10.42 GHz. In the X band, the concave mode achieves -22.6 dB average reflectivity. In Fig. 7 (f), the convex mode red shifts the reflectivity peak and enhances peak II, reaching $RL_{\min}$ of -57 dB at 12.94 GHz. In the entire K_u band, the reflectivity of the absorber remains below -20 dB, achieving average reflectivity of -24.1 dB.

Therefore, the integration of CNT-FSS device on the metastructure can enhance the absorption peak at specific frequency bands, thereby enabling a shift in the frequency band corresponding to perfect absorption peak. As a consequence, the absorber can reconfigure to cover both the X and K_u bands with an average RL lower than -20 dB, achieving ultra broadband near-perfect absorption characteristics. Additionally, the shift in the perfect absorption peak corresponding to different frequency bands can provide insights for the development of highly sensitive sensors. In addition, the gray curve in Fig. 7 (b) represents the reflectivity when the FSS is suspended on the structure. In this condition, the absorber generate EAB in the range of 5.8–18 GHz, demonstrating consistent absorbing performance with the CMS and offering a means to switch off the FSS function. the metastructure can achieve the activation/deactivation of the FSS function, and facilitate a shift in the frequency band corresponding to perfect absorption peak.

4. Discussion

4.1. Discussion

The proposed cement-based reconfigurable metastructure, comprising a mechanically stable CMS and a reconfigurable CNT-FSS

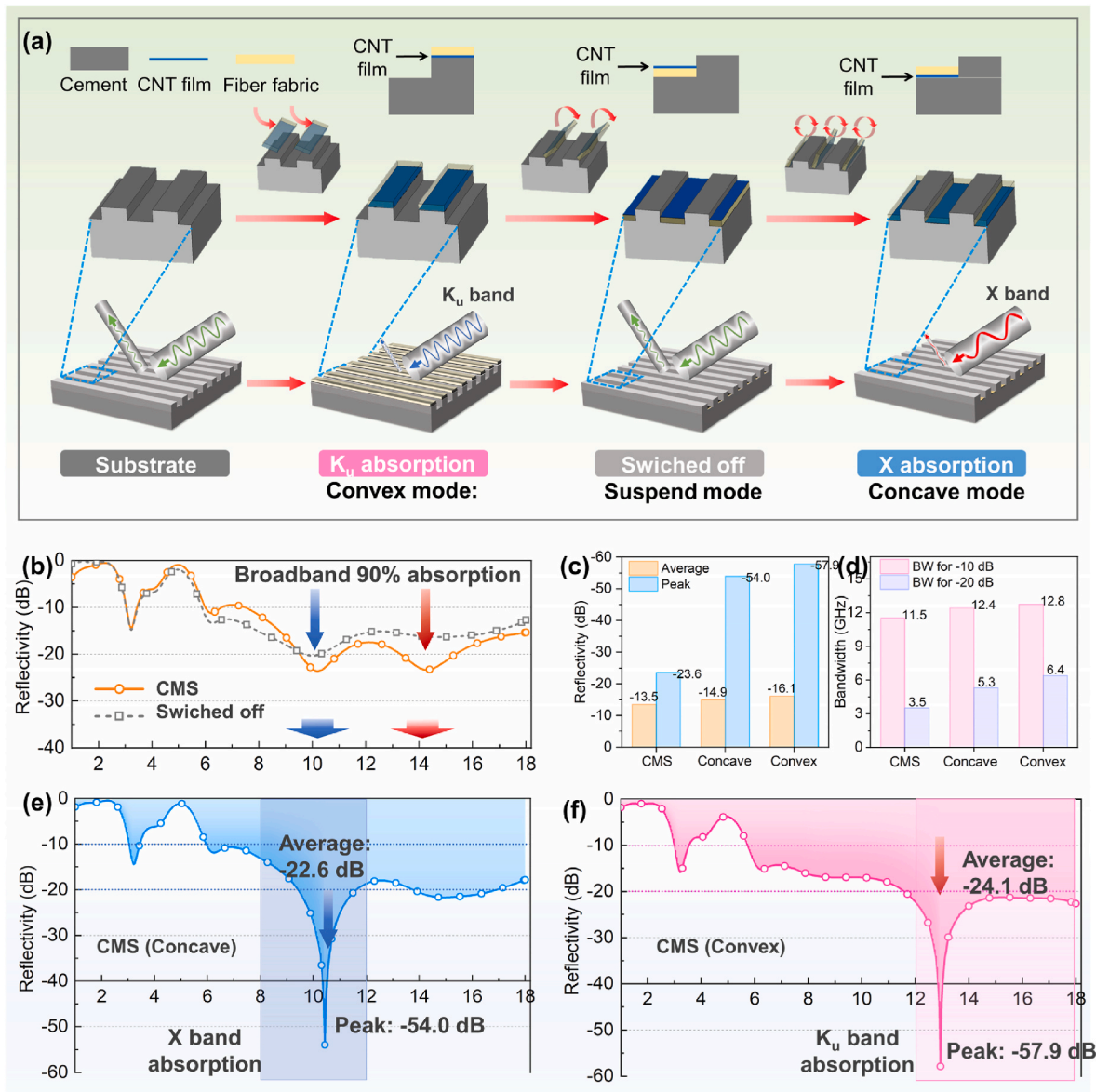


Fig. 7. (a) The schematic diagram of three structural forms combining CNT-FSS with CMS. (b) Reflectivity results of the CMS and the suspend mode. (c) Statistics of average reflectivity and peak reflectivity of the three structures. (d) The -10 dB and -20 dB bandwidths of the three structures. Reflectivity results of the (e) CMS (Convex) and (f) CMS (Concave).

device, offers a novel solution to the demand for deformable and reconfigurable absorbers while overcoming issues of mechanical stability and practical integration in engineering applications. Table 3 compares the EM wave absorbing performance of competitive absorbers and the reconfigurable metastructure in this work. It can be found that the absorbing composites can achieve low $RL_{\min}$ and broad EAB at thin thickness. Furthermore, by strategically implementing metastructure and multilayer structure designs, the EAB can be further extended. In recent years, there has been a growing interest in the development and utilization of tunable wave absorbers, which offer the advantage of adaptability to various operational conditions. For example, Zhao et al. designed a hierarchical porous molybdenum carbide matrix, and superior RL and EAB in both X and K_u bands can be achieved through thickness modulation [57]. Gao et al. proposed a method to selectively grow CNT cluster arrays on carbon fiber surfaces for tunable microwave absorption. The CF/CNT cluster array composites exhibit excellent wave absorption performance with 4.4 GHz EAB, extendable to the full frequency range from 2 to 18 GHz through metastructure design [58]. Unlike traditional approaches that add absorbing agents to composites,

this study adopts a novel method combining cement-based metastructures with CNT-FSS to achieve broadband absorption performance, low $RL_{\min}$, and tunable absorption performance.

In the metastructure proposed in this study, the CNT-FSS device and the CMS substrate jointly achieve frequency selection characteristics. The CMS substrate can effectively modulate and aggregate EM waves of various frequencies to different targeting positions. The CNT-FSS device, fabricated by coating CNT solution onto the fiber fabric, forms an impedance gradient structure with enhanced loss characteristics. The device is strategically integrated at the targeting positions of the CMS to effectively dissipate aggregated EM waves at specific frequencies, thereby achieving frequency selective absorption properties. Moreover, the CNT-FSS device can provide convenient deformation and reconfiguration due to its flexibility and mechanical properties. Specifically, in the experimental tests, the structural transformation is achieved by flipping the CNT-FSS device, as shown in Fig. 1 (c). In practical applications, it is possible to design mechanical hinges on the device or integrate origami structures with the metastructure to guarantee the structure more responsive to external stimuli.

Table 3

The EM wave absorbing performance of competitive absorbers and the reconfigurable metastructure in this work.

Structure form	Material	Test range (GHz)	Thickness (mm)	Agent content	$RL_{\min}$ (dB)	EAB (GHz)	Tunable property	Refs.
Composite	SnS/SnS ₂ /SnO ₂ /CF	2–18	2	50 wt%	−46.7	5.3	No	[50]
Composite	NiCo ₂ Se ₄ spinel	2–18	2.4	5 wt%	−53.9	8.5	No	[59]
Composite	Micro-sized hexapod-like CuS/Cu ₉ S ₅	2–18	2.3	25 wt%	−59.4	7.4	No	[60]
Composite	Cr ₂	2–18	1.4	40 wt%	−52.7	4.1	No	[61]
Composite	MIL-88A@Ni–Fe LDH@Ti ₃ C ₂ T _x -1.0	2–18	1.8	–	−46.7	5.1	No	[62]
Composite	Ni/Co/CrN/CNTs-CF	2–18	2.1	10 %	−56.2	5.8	No	[63]
Composite	MXene/ZrO ₂ /W–C	2–18	3.8	4.8 wt%	~−30	11.1	No	[51]
Fabric	CF/MXene	2–18	2.4	80 wt%	−57.1	7.7	No	[64]
2 layered structure	Foam cement-based material	2–18	30	0.12 wt% CNT	−47.3	14.0	No	[12]
3D porous periodic MM	Foam cement-based material	2–18	40	0.5 wt% CF	−56.6	14.8	No	[17]
3 layered Honeycomb composite	CNTs/CB/PU	2–18	15	4.7 %	−26.4	15.5	No	[56]
2 layered Honeycomb composite	RGO/nonwoven absorbent	2–18	10	4.7 %	~−23	16	No	[53]
3D porous structure	MS/MC/PNC-2 composite	2–18	2.4	45.5 wt%	−55.3	7.6	Yes	[57]
Composite	CF/CNTs cluster array	2–18	1.4	5 wt%	~−22	4.4	Yes	[58]
Metastructure			45	4–6 wt%	~−27	16		
Composite	NiSe/ZnSe/MHCFs	2–18	2.6	45 %	−60.6	7.4	Yes	[52]
Metastructure + CNT-FSS	CNT/Cement	1–18	25	0 %	−57.9	12.8	Yes	This work

4.2. Outlook

On the one hand, the cementitious material used in the study is ultra high toughness cementitious composite (UHTCC), which is a fiber-reinforced cementitious material with quasi-strain hardening and a fine multiple cracking phenomenon [65,66]. Its tensile strain reaches 3 %–10 % (300–1000 times that of ordinary concrete) [67,68]. In our previous work, this material has effectively combined with CNTs and achieved good mechanical and absorbing properties [48,69,70]. The UHTCC has been used in various infrastructure such as building construction, roads, and bridges, ensuring the stability and durability of the metastructure absorber in this work, promoting its practical application in infrastructure engineering. On the other hand, the experimental results demonstrate the capability of the metastructure to dynamically regulate its absorption across different frequency bands, showing potential for broadband absorption and offering impressive average and peak reflectivity in tunable frequency bands.

Based on the modulation and aggregation mechanisms of EM energy by the CMS, the dynamic absorption performance of the metastructure can be extended to broader frequency ranges. Moreover, by leveraging the method of integrating the CMS and the CNT-FSS device, researchers can apply this method to dynamically regulate additional properties such as absorption, reflection, and transmission. Future research could focus on addressing the scalability of this technology for industrial applications, such as EM stealth, EM interference shielding, dynamic EM energy harvesting, and highly sensitive sensors.

5. Conclusion

This study proposes a reconfigurable cementitious metastructure, consisting of a mechanically stable CMS and reconfigurable CNT-FSS device, enabling tunable perfect EM absorption. The 3D CMS, serving as the substrate of the CNT-FSS, exhibits significantly improved absorption performance compared to the cementitious slab absorber for both TE and TM polarizations. CST simulations are employed to calculate the distribution of electric and magnetic fields, as well as power loss in CMS. Then this study explores the integration of CMS with CNT-FSS to modulate EM absorption, thereby proposing a class of reconfigurable metastructures for dynamic control of absorption performance. As a proof of concept, a flexible CNT-FSS device is designed and fabricated, capable of forming an impedance gradient structure with enhanced loss

characteristics and modulating absorption performance through flipping the device. Experimental results demonstrate that the absorption performance of the metastructure switches between broadband absorption (reflectivity below −10 dB in 5.8–18 GHz), perfect X band absorption (−22.6 dB average reflectivity and −54 dB $RL_{\min}$), and perfect Ku band absorption (−24.1 dB average reflectivity and −57.9 dB $RL_{\min}$). The absorbing structure not only significantly enhances and enriches the performance of the cementitious absorbers, but also provides a novel approach to actively optimize absorption performance and adjust operational frequency bands for diverse EM scenarios including broadband wave absorption, EM energy harvesting, and sensors.

CRedit authorship contribution statement

Xiaoran Wang: Writing – original draft, Methodology, Investigation, Funding acquisition, Data curation. **Qinghua Li:** Writing – review & editing, Supervision, Resources, Funding acquisition. **Haixin Lai:** Investigation, Data curation. **Shilang Xu:** Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.carbon.2025.120018>.

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